

HDOC Day_28 Programming in C

Problem_1

Write a C program to:

a) Read the number of rows and columns from the user. Ensure that both values are positive. If not, repeat to get

the desired input(s).

b) Declare a variable-length 2D integer array using the given dimensions.

c) Read the array elements column by column using nested loops, with the outer loop iterating over columns and

the inner loop over rows.

d) Print the elements in row-major order using a single loop and a linear index

CODE

```
#include <stdio.h>

int main()
{ int x=0;

  int r,c;

  while(x==0)

  {

    printf("enter no for row and column\n");

    scanf("%d%d",&r,&c);

    if(r>0 && c>0)

      break;

  }printf("enter elements of array\n");

  int arr[r][c];

  int length=r*c;

  for(int i=0;i<c;i++)

  {

    for(int j=0;j<r;j++)
```

```

    {
        scanf("%d",&arr[j][i]);
    }
}
for(int i=0;i<length;i++)
{
    int n=i/c;
    int m=i%c;
    printf("%d\t",arr[n][m]);
}
return 0;
}

```

Problem_2

Write a C program that reads elements into a variable-length 2D array, copies them into a 1D array in row-major order, and displays both arrays

CODE

```

#include <stdio.h>

int main()
{ int x=0;
  int r,c;
  while(x==0)
  {
      printf("enter no for row and column\n");
      scanf("%d%d",&r,&c);
      if(r>0 && c>0)
          break;
  }printf("enter elements of array\n");
  int arr[r][c];

```

```
int length=r*c;
for(int i=0;i<c;i++)
{
    for(int j=0;j<r;j++)
    {
        scanf("%d",&arr[j][i]);
    }
}

int sz=r*c;
int arr2[sz];
int count=0;
for(int i=0;i<r;i++)
{
    for(int j=0;j<c;j++)
    {
        arr2[count]=arr[i][j];
        count++;
    }
}

printf("1d array is\n");
for(int i=0;i<sz;i++)
{
    printf("%d\t",arr2[i]);
}

printf("\n2nd array is\n");
for(int i=0;i<r;i++)
{
    for(int j=0;j<c;j++)
```

```
{  
    printf("%d\t",arr[i][j]);  
}  
    printf("\n");  
}  
return 0;  
}
```